

上海交通大学研究生课程教学大纲

Syllabus for SJTU Graduate Courses

课程基本信息 / Basic information	内容 / Details
课程名称 / Course name	云计算 / 云计算和人工智能前沿进展
开课学期 / Semester	2026—2027 学年秋季学期 / Fall 2026
课程层次 / Level	研究生课程 / Graduate course
教学资料 / Materials	两个教学班共用课程内容与教学材料 / Shared content and materials for both classes

课程简介与教学方式 / Description and teaching methods

项目 / Item	内容 / Content
课程简介（中文）	本课程围绕云计算与人工智能的交叉应用，讲授云计算与机器学习基础、联邦学习、无服务器计算及量子计算的基本概念、关键技术与研究进展。课程结合理论讲授、案例分析和编程实践，培养学生分析计算系统、设计实验和解释研究结果的能力。 学生通过课程学习，理解云计算的基本服务模式及其对机器学习训练与推理的支持方式；掌握联邦学习的本地训练、参数通信与聚合流程，分析数据分布和计算资源差异的影响；理解无服务器计算的执行模型、典型应用及性能开销；了解量子计算与量子机器学习的基础方法和应用边界；具备阅读相关文献、实现实验方案、分析结果和汇报研究工作的能力。
Course description (English)	The course examines the intersection of cloud computing and artificial intelligence. It covers foundations of cloud computing and machine learning, federated learning, serverless computing, and quantum computing. Lectures, case studies and programming activities develop students' ability to analyse computing systems, design experiments and interpret research results. Students learn how cloud services support model training and inference; understand local training, communication and aggregation in federated learning; analyse the effects of data and resource heterogeneity; understand the execution model and performance costs of serverless computing; and examine fundamental methods and application boundaries of quantum machine learning. Students develop skills in reading research literature, implementing experiments and presenting their findings.
教学方式（中文）	课程采用理论讲授、案例讨论、项目实践、企业专家授课和项目汇报相结合的教学方式。联邦学习实践围绕语言模型微调项目开展；无服务器计算与量子机器学习由教师或助教演示，学生观看并参与案例讨论。教学资料通过 Canvas 平台提供。
Teaching methods (English)	Teaching combines lectures, case discussions, project practice, three industry-expert lectures and a project presentation. Federated learning practice supports the language-model fine-tuning project. The instructor or teaching assistant demonstrates serverless computing and quantum machine learning cases; students observe and discuss the results. These two demonstration sessions require no student operation or submission and carry no separate grade. Learning materials are provided through Canvas.

教学大纲（中文）

课程按 16 次教学安排组织，实际授课按校历执行。项目于第 13 教学周统一提交，教学周与课次分别按课表核对。表格采用课次标识教学安排。(2025 年模板列示 2 学分、32 学时。)

模块	教学内容（要点）	教学目标	课次	教学形式	作业及考核要求
基础	云计算与机器学习概论	学生理解云计算服务模式及机器学习基本流程。	1	理论讲授	课程要求与考核说明
联邦	联邦学习概论	学生理解本地训练、通信与参数聚合流程。	2	理论讲授	综合项目理论基础
联邦	联邦学习与语言模型微调实践	学生掌握 Notebook 操作和微调基线的实现方法。	3	项目实践指导	发布一次综合项目；阶段一指导
联邦	联邦学习前沿	学生分析通信效率、隐私保护与数据异质性问题。	4	理论讲授	综合项目方法分析
专题	企业专家授课（一）	学生理解云计算与人工智能的产业应用。	5	企业专家授课	课堂学习与案例交流
无服务器	无服务器计算概论	学生理解函数执行、资源管理与服务调用。	6	理论讲授	课堂学习
无服务器	无服务器计算案例演示	学生通过演示分析函数调用、任务分块与性能开销。	7	教师或助教演示	观看与讨论；无操作、无提交、无独立评分
无服务器	无服务器计算前沿	学生分析无服务器计算研究问题及模型推理应用。	8	理论讲授	课堂学习
专题	企业专家授课（二）	学生分析实际系统中的技术选型与应用约束。	9	企业专家授课	课堂学习与案例交流
联邦	联邦微调实验设计与分析	学生设计对照实验并分析数据分布、规模与资源开销。	10	项目实践指导	同一项目阶段二、三指导
量子	量子计算与量子机器学习基础	学生理解量子态、测量与量子机器学习基本方法。	11	理论讲授	课堂学习
量子	量子机器学习案例演示	学生通过演示解释测量采样、分类结果与实验条件。	12	教师或助教演示	观看与讨论；无操作、无提交、无独立评分
量子	量子计算前沿与交叉应用	学生认识量子计算的交叉应用与研究局限。	13	理论讲授	课堂学习
专题	企业专家授课（三）	学生分析行业前沿案例及技术发展问题。	14	企业专家授课	课堂学习与案例交流
汇报	联邦微调综合项目汇报	学生阐述项目方法、实现、结果与局限。	15	项目汇报与个人问答	同一项目汇报，占总评 20%
复习	课程综合复习与机动	学生梳理各模块联系及核心方法。	16	综合复习	课程总结与机动安排

Syllabus (English)

The course is organised into 16 sessions, subject to the academic calendar. The integrated project is submitted once in teaching week 13; session numbers and teaching weeks are mapped to the class timetable.

Module	Teaching content	Learning objectives	No.	Teaching method	Assignment and assessment
Basics	Cloud computing and machine learning foundations	Students explain cloud service models and the basic machine learning workflow.	1	Lecture	Course and assessment overview
FL	Introduction to federated learning	Students explain local training, communication and parameter aggregation.	2	Lecture	Theory for the integrated project
FL	Federated learning and language model fine-tuning practice	Students use notebooks and implement a fine-tuning baseline.	3	Project guidance	One integrated project released; stage 1 guidance
FL	Frontiers in federated learning	Students analyse communication efficiency, privacy and data heterogeneity.	4	Lecture	Methods for the integrated project
Industry	Industry expert lecture I	Students explain industrial applications of cloud computing and AI.	5	Industry lecture	Classroom learning and case discussion
FaaS	Introduction to serverless computing	Students explain function execution, resource management and service invocation.	6	Lecture	Classroom learning
FaaS	Serverless computing case demonstration	Students analyse function invocation, task partitioning and performance overhead through demonstrations.	7	Instructor or TA demonstration	Observation and discussion; no student operation, submission or separate grade
FaaS	Frontiers in serverless computing	Students analyse serverless research problems and model inference applications.	8	Lecture	Classroom learning
Industry	Industry expert lecture II	Students analyse technology choices and practical constraints in deployed systems.	9	Industry lecture	Classroom learning and case discussion
FL	Experimental design and analysis for federated fine-tuning	Students design controlled comparisons and analyse data distribution, scale and resource costs.	10	Project guidance	Stages 2 and 3 guidance for the same project
Quantum	Quantum computing and quantum machine learning foundations	Students explain quantum states, measurement and basic quantum machine learning methods.	11	Lecture	Classroom learning
Quantum	Quantum machine learning case demonstration	Students interpret measurement sampling, classification results and experimental conditions through demonstrations.	12	Instructor or TA demonstration	Observation and discussion; no student operation, submission or separate grade
Quantum	Frontiers and interdisciplinary applications of quantum computing	Students identify interdisciplinary applications and research limitations of quantum computing.	13	Lecture	Classroom learning
Industry	Industry expert lecture III	Students analyse current industry cases and issues in technology development.	14	Industry lecture	Classroom learning and case discussion

Module	Teaching content	Learning objectives	No.	Teaching method	Assignment and assessment
Project	Integrated federated fine-tuning project presentation	Students present project methods, implementation, results and limitations.	15	Presentation and individual questions	Presentation of the same project; 20% of the final grade
Review	Course review and flexible session	Students synthesise connections and core methods across course modules.	16	Review	Course summary and flexible scheduling

课程要求与资源 / Requirements and resources

项目 / Item	内容 / Content
课程要求（中文）	课程设置一次联邦微调综合项目，于第 13 教学周提交实验报告、Notebook 与辅助源码、环境与运行说明、配置、必要日志及汇报材料。总评由考勤 10%、项目实验与材料 70%、项目汇报 20% 组成。项目三个阶段分别为基础训练与基线建立 20 分、联邦微调与对照分析 30 分、规模扩展与优化 20 分。阶段分数直接相加。无服务器计算与量子机器学习两次案例课由教师或助教演示，学生观看并参与讨论，不安排学生操作、材料提交或独立评分。
Requirements (English)	The course includes one integrated federated fine-tuning project, submitted in teaching week 13 with a report, notebook, supporting code, environment and execution instructions, configurations, required logs and presentation materials. The final grade comprises attendance (10%), project experiments and materials (70%), and the project presentation (20%). The three project stages are baseline training (20 points), federated fine-tuning and comparisons (30 points), and scaling and optimisation (20 points). Stage scores are added directly. Serverless and quantum machine learning case sessions are demonstrated by the instructor or teaching assistant. Students observe and discuss the results; these sessions require no student operation or submission and have no separate grade.
评分依据（中文）	项目材料按代码与可复现性 25 分、实验设计与结果 30 分、报告分析 15 分归集；汇报按方法理解、实现掌握、结果与局限分析、组织与证据展示四项评价，各 5 分。课程组按统一量表记录分项得分与证据并实施复核。
Assessment criteria (English)	Project materials comprise implementation and reproducibility (25 points), experimental design and results (30 points), and analysis (15 points). The presentation assesses method understanding, implementation knowledge, interpretation of results and limitations, and organisation and evidence, with 5 points each. Component scores and supporting evidence are recorded and reviewed using a common rubric.
课程资源（中文）	授课讲义、三阶段项目说明、Notebook 示例、教师演示与课程评分细则通过 Canvas 提供。参考文献结合各专题列示。
Resources (English)	Lecture slides, the three-stage project specification, notebook examples, instructor demonstrations and assessment rubrics are provided through Canvas. Research references accompany each topic.